Do Better Port-Pressure Intervals Change Cargo-Aware Voyage Decisions? A Monte Carlo Study for Indian Ports

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***Abstract*—Voyage planners need to know whether better uncertainty estimates change route and speed choices when delays carry cargo costs. Across 12 Indian ports, gradient boosting with traffic history and calendar features had the lowest benchmark mean absolute error (6.25 index points). Causal HMM regime features did not improve the GBM. At nominal 80% in the forecast-to-decision experiment, static conformal recalibration increased coverage from 72.1% to 75.1%. Monthly rolling calibration improved on the base forecast but scored worse than static calibration. On synthetic voyages with illustrative cargo contracts, expected-cost selection cut mean cost regret by USD 9,121 versus the time-based incumbent while increasing mean time regret by 70.75 h. Both risk-averse rules raised mean time and cost regret but cut regrets above USD 10,000 from 20.9% to 4–6%, with one shared USD 759,546 loss. In a synthetic Suez case, shared scrambled Sobol draws reduced the cost-gap standard error by a factor of 11.7 versus independent draws. Choices were insensitive to recalibration because changes in expected berth-wait cost were small relative to the decision margins. In the stress grid, the predictive CVaR rule first switched at least 1% of choices at triple interval width with the original wait slope. The expected-cost rule stayed fixed across the grid.**

***Index Terms*—Port congestion proxies, conformal prediction, cargo risk, decision support, common random numbers, quasi-Monte Carlo.**

I. Introduction

Suez and Red Sea disruptions can force longer voyages, and the Strait of Hormuz matters for cargo flows through the Persian Gulf [1]. Automatic Identification System (AIS) port calls provide a proxy for activity when verified berth and anchorage outcomes are unavailable [2]. Route planning also requires assumptions about how that uncertainty affects arrival times, delivery penalties, and insured exposures. We study whether improving forecast intervals changes a decision once these cargo costs are priced.

Our research questions are whether port-pressure forecasts are calibrated and recalibration helps (RQ1), whether cargo-aware objectives change choices and at what time and tail cost (RQ2), and whether better calibration propagates to decisions (RQ3).

- A chronological forecast benchmark that removes a look-ahead feature and compares static and rolling calibration.
- Shared random numbers and scrambled Sobol sampling reduced the cost-gap standard error by a factor of 11.7 in a synthetic Suez fixture.
- A cargo-aware objective evaluated against time-based choices, with a paired tail audit.
- A forecast-to-decision test in which improved empirical coverage left the synthetic route and speed choices unchanged.

We review related work, describe the forecast and decision methods, and report the results by research question before discussing their use.

II. Related Work

AIS trajectories support measurement of anchorage and service time [3]. Tree-based models classify congestion at Busan [4], and terminal optimisation studies berth and crane allocation [5]. We use a continuous relative-call index as an activity proxy.

Gradient boosting supports quantile forecasts [6], while conformalized quantile regression adjusts interval endpoints on held-out data [7]. Adaptive conformal inference and EnbPI address time-dependent calibration in different ways [8], [9]. We compare static and monthly rolling corrections. Temporal Fusion Transformers can encode covariates [10]. We test a hidden Markov model (HMM) with empirical dwell-time estimates [11], [12].

Chance constraints, conditional value-at-risk (CVaR) and minimax regret rank options under different risk criteria [13]–[15]. Common random numbers pair option outcomes [16]. Sobol nets distribute points across the integration domain and independent randomisations estimate numerical error [17]–[19]. Liner-shipping research studies routing, speed, fuel use, and the value of time [20]–[22]. Here the ranking prices delivery terms for individual consignments. The separate berth-allocation simulator tests masked tabular Q-learning [23].

III. System Overview

Fig. 1 and Table I show the system and its evidence. Source receipts preserve observation and availability times. Company reviews route and cost options, Port reviews local constraints, and Vessel receives the advisory and its provenance. Authorised roles retain clearance and execution. Source specialists route requests by rules to deterministic, permission-scoped tools. They call no language model.

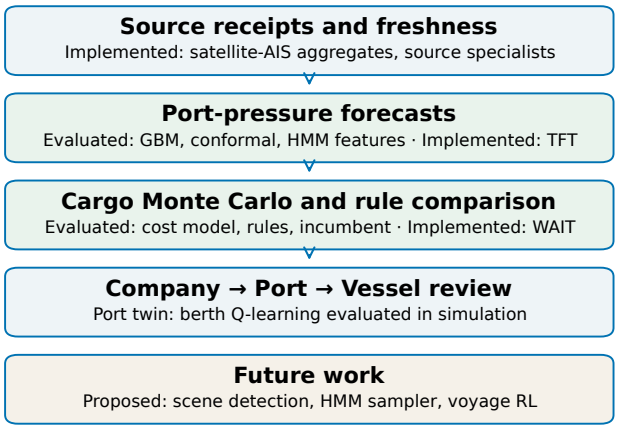


Fig. 1. Evaluated, implemented and proposed components in the advisory workflow.

TABLE I
SYSTEM COMPONENTS AND EVIDENCE.

Component	Role	Status	Evidence
GBM, conformal	Forecast	Evaluated	Chronological holdout
HMM regimes	Features	Evaluated	Matched forecast ablation
TFT	Forecast	Implemented	Optional training path
Cost, R1–R5, incumbent	Decisions	Evaluated	Synthetic case splits
WAIT diagnostic	Decisions	Implemented	Value of information
Masked tabular Q-learning	Berths	Evaluated in simulation	-211.99 vs. -179.31 test reward (lookahead, 100 episodes). Rejected.
Source specialists	Evidence	Implemented	Rule routing, scoped tools and provenance
Satellite-AIS aggregates	Activity	Implemented	Daily port calls
Scene SAR/optical	Detection	Proposed	Future work
HMM-driven Monte Carlo	Futures	Proposed	Future work
Voyage RL	Decisions	Proposed	Future work

IV. FORECASTING METHOD

A. Target, Features, and Split

For port p and date t, let $a_{p,t}$ be the trailing 7-day average of AIS-derived vessel calls and let $b_{p,t}$ be the expanding mean of those averages strictly before t [2]. The pressure target is

$$y_{p,t} = \mathrm{clip}_{[0,100]}\left(50\,\frac{a_{p,t}}{b_{p,t}}\right). \tag{1}$$

A value of 50 represents the port's own historical baseline. The denominator requires 14 prior values. For earlier dates the pipeline fills the index with the first available index value. All scored targets occur after that initial period.

Traffic inputs include the index, estimated throughput, and a delay proxy. They use origin-day values, lags at 1, 3, and 7 days, and trailing means. Learned models also use static port descriptors and horizon. Calendar features describe the target date. The archived field `weather_forward_load` matched the mean of the next 5 composite values on every complete window. We removed it along with other unverified contextual features. The restricted benchmark compares history alone, history plus calendar, and history plus calendar plus 7-day-lagged ERA5 weather [24]. The lag is a retrospective availability assumption.

The comparison uses 5 expanding-window folds. Model fitting admits only targets available by the cutoff under the panel's date assumption. Ensemble weights use earlier out-of-fold predictions and interval adjustment uses a separate chronological block. We exclude training targets that cross the fold cutoff. The final reporting period reuses previously examined dates. Persistence repeats the origin value, the weekly-lag rule repeats its value from a week earlier, and ridge uses horizon-specific residual bands. The adaptive ensemble combines persistence, boosting, and ridge.

B. Quantiles and Recalibration

Histogram gradient boosting (GBM) learns the residual from the origin index for each quantile τ with the pinball loss $\rho_\tau(e) = \max(\tau e, (\tau-1)e)$. For each lead and coverage level $1-\alpha$, we compute

$$s_i = \max(\ell_i - y_i,\ y_i - u_i,\ 0)$$

on validation rows. The correction $\widehat{s}$ is the order statistic at rank $\lceil (n+1)(1-\alpha) \rceil$ [7]. We expand the raw band to $[\ell - \widehat{s}, u + \widehat{s}]$, clip it to the index domain, and enlarge wider bands as needed to preserve nesting. The monthly rolling variant starts with the static corrections and recomputes them by lead and nominal level using the preceding 48 calendar days of forecast origins and targets dated before each update. It freezes offsets within each month and keeps the GBM fixed.

C. Causal regime ablation

We fit an HMM with empirical dwell-time estimates to daily pressure in the training block through the early cutoff. Forward filtering uses each port's history through the origin. The ablation adds state probabilities, days in the most probable state, expected remaining days and escalation risk to the same GBM. Both arms use the same rows and split conformal procedure. We resample paired 10-day blocks of origin dates, keeping all ports and leads together. Success requires both pooled MAE and static 80% interval score to improve with 95% intervals excluding zero.

V. DECISION METHOD

A. Consignments and Cost

Each consignment declares a deadline, priority, dwell and detention free time, value and carrying rate, and, where relevant, remaining shelf life, power tolerance, and insured route exposure. For option j, let A_{ij} be delivery time, D_{ij} positive delay relative to plan, B_{ij} dwell, T_{ij} detention, and U_{ij} hours without power. With $x_+ = \max(x, 0)$ and deadlines

and tolerances denoted by lower case variables, the voyage loss is

$$
\begin{aligned}
L_j &= C_j^{\text{voyage}} + \sum_i L_{ij}, \\
L_{ij} &= C_i^{\text{late}}((A_{ij} - d_i)_+) + \frac{r_i^{\text{dem}}(B_{ij} - b_i)_+}{H_d} \\
&\quad + \frac{r_i^{\text{det}}(T_{ij} - t_i)_+}{H_d} + \frac{V_i r_i^{\text{annual}} D_{ij}}{H_y} \\
&\quad + V_i \mathbb{I}\{U_{ij} > u_i \vee A_{ij} > s_i\} + C_{ij}^{\text{ins}}, \\
C_{ij}^{\text{ins}} &= V_i^{\text{ins}} \left(r_i^{\text{base}} + \sum_{a \in \mathcal{A}_j} r_{ia}^{\text{war}} \right) + C_{ij}^{\text{ret}} + C_{ij}^{\text{GA}}.
\end{aligned}
\tag{2}
$$

$H_d = 24$ and $H_y = 8760$ convert hourly clocks to days and years. Late penalties can charge once or at a daily rate. $\mathcal{A}_j$ identifies the applicable war-risk areas. Retention and general-average cost require declared event and coverage terms. An illustrative default uses 1.10 times cost, insurance, and freight value. Voyage charges and fuel enter once. Missing terms are declared and excluded. A known subtotal is comparable only when the same components and basis apply across options. Cost rollups preserve the shared scenario draw.

B. Sampling and Stopping

We transform shared uniforms into closure, weather, port pressure, handling, fuel and available exchange rates using inverse CDFs and a declared Gaussian copula. The main comparison uses fallback wait knots of 0, 3 and 12 h at the 10th, 50th and 90th percentiles. The forecast overlay substitutes the pressure quantiles at the same probabilities. Both inverse CDFs interpolate between knots and hold draws outside the outer probabilities at the nearest endpoint, creating mass at both ends. The sensitivity run extends the adjacent segments linearly, floors wait at zero and bounds pressure to the index scale, keeping Sobol uniforms and realised outcomes fixed.

We estimate the standard error (SE) of the mean cost gap by dividing the standard deviation of independent scramble means by the square root of the number of scrambles. The adaptive run doubles complete prefixes until the top-two order persists and the paired SE meets the threshold, or it reaches the cap. This criterion controls precision for the mean gap.

C. Decision Rules and Promotion

Let j_k be keep, $\mu_j = \mathbb{E}L_j$, p_j the probability of missing any priority-1 deadline, and V_α value-at-risk. For continuous loss, $\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq V_\alpha(L)]$. For mass at the cutoff we use

$$
\text{CVaR}_\alpha(L) = \min_z \left(z + \frac{\mathbb{E}[(L - z)_+]}{1 - \alpha} \right). \tag{3}
$$

Let $v_j = \text{CVaR}_\alpha(L_j)$. The rules share feasible options and sample outcomes.

- R1 Minimise μ_j.
- R2 Minimise μ_j subject to $p_j \leq \epsilon$. Abstain if there is no eligible option with a declared probability.
- R3 Minimise $\text{CVaR}_{0.95}(L_j)$.
- R4 Minimise $\max_\omega[L_j(\omega) - \min_i L_i(\omega)]$.
- R5 Among interventions with $\Pr(L_j < L_{j_k}) \geq q$, $v_j \leq v_{j_k}$, and $p_j \leq \epsilon$, minimise μ_j. Keep if no intervention qualifies.

The robust incumbent minimises the worst-case time regret over four closure horizons: FIZZLE, SHORT, BASE and LONG (no closure, half the remaining claimed duration, that duration, and twice that duration). It compares each option to the quickest feasible option in each horizon and may keep or defer under its intervention and evidence checks. The time-only comparator minimises expected delivery time. Before evaluating R5, we set $q = 0.90$ and $\epsilon = 0.10$ in the tuning grid. The promotion test required sufficient case coverage, limited vessel time regret, enough beneficial vessel interventions, and a paired cargo cost-tail check on validation and test. Coverage counts all offered cases, including cases without a comparable realised outcome.

VI. Experimental Setup

A. Forecast and Decision Samples

The frozen PortWatch panel contains 4,740 port-days at 12 Indian ports from 2025-09-03 to 2026-10-02. We test 1–10-day leads. The forecast overlay uses a GBM trained through 2026-07-05. Calibration origins start 2026-07-06 and their retained targets end 2026-08-22. Test origins span 2026-08-23 to 2026-10-01 with targets through 2026-10-02. Each model contributes 4,260 test forecasts. Forecast metrics are median MAE, root mean squared error (RMSE), empirical interval coverage, and mean interval width.

The test design offers 200 synthetic vessel cases and 200 synthetic cargo cases, with 190 and 182 scored by R1. The vessel cost is an illustrative USD 250/delay-hour conversion. Cargo has distinct seeded consignment mixes and hidden futures. We score time and cost regret against the best feasible realised choice for each objective. We report mean, 90th percentile, and maximum cost regret, realised cost CVaR, and the share exceeding USD 10,000 regret. Capture requires a selected intervention to improve on keep when one could do so. We measure unnecessary interventions among interventions with a comparable realised outcome. Paired 95% percentile bootstrap intervals [25] use 10,000 resamples. Two-sided paired Wilcoxon tests use the Pratt zero convention and are descriptive.

B. Forecast-to-Decision Experiment

The forecast overlay assigns rows without replacement by lead time to frozen cargo cases and illustrative fixtures. Forecast ports and dates are unrelated to the synthetic voyages. Each pair uses the original and calibrated nominal 80% endpoints and the same median, option set, cargo terms, and Sobol uniforms. A declared local wait function

$$
B^{\text{wait}}(Y) = \max\{0,\ 3 + 0.36(Y - \hat{q}_{\text{med}})\}
$$

maps sampled pressure Y to hours before applying the option's berth exposure. This transfer is illustrative. The original

TABLE II
FORECAST ACCURACY AND NOMINAL 80% COVERAGE. THE MATCHED EARLIER-CUTOFF ROWS SHOW RAW BANDS. THE LATER BENCHMARK INCLUDES ITS CALIBRATED ENSEMBLE. ERRORS AND WIDTH ARE IN INDEX POINTS.

Model	MAE	RMSE	Cov. (%)	Width
GBM history + calendar	6.25	8.48	70.1	17.00
GBM + lagged weather	6.28	8.54	73.2	17.38
Adaptive ensemble	6.36	8.70	87.3	23.37
GBM history only	6.40	8.63	84.0	21.54
Ridge	6.49	8.46	85.8	23.36
Persistence	6.67	9.10	88.4	26.13
Weekly lag	8.42	11.29	–	–
Matched earlier-cutoff ablation				
GBM history + calendar	6.40	8.74	72.1	18.03
GBM + causal regimes	6.43	8.78	72.1	18.11

decision runs assumed a berth-wait distribution. The overlay retains the original hidden outcomes for cargo and uses 4,096 independent reference futures per illustrative fixture. The Ever Given reference fixes reopening to the recorded timeline while retaining illustrative inputs. The overlay uses 4 scrambles of 512 futures each for each cargo case and 8 of 2,048 per fixture. R2 resolves abstentions to keep in the overlay.

C. Reproducibility and Runtime

The experiments use seed 20261112. Run `python` with `scripts/paper_experiments.py` and `--all` for the decision runs, then `--e7` with the pinned forecasting interpreter. The manifest records inputs, outputs, and timings. The decision benchmark ran on Windows with an Intel CPU and 24 logical processors in 209.87 s. The forecast overlay took 61.60 s. The `--regime` mode reproduces the registered forecast ablation.

VII. RESULTS

A. RQ1: Forecast accuracy

Table II shows the chronological benchmark. GBM history plus calendar leads in pooled MAE, while ridge has the lowest RMSE. Persistence has lower day-1 MAE, and the GBM is better at subsequent horizons. Adding lagged weather increases MAE slightly. The supplement plots errors by lead.

On the matched earlier-cutoff rows, regime features worsened MAE by 0.03 points (95% CI -0.03 to 0.12) and the static 80% interval score by 0.63 (0.20 to 1.14). The MAE interval includes zero, and the ablation missed the joint success criterion. The supplement reports all leads, threshold-exceedance scores and the spell-duration diagnostic.

B. RQ1: Forecast interval calibration

Table II reports the later-cutoff benchmark fit (MAE 6.25). The forecast overlay refits the same GBM specification at an earlier cutoff to reserve a clean calibration block (MAE 6.40) and holds that refit's median fixed. Recalibration widens intervals and increases coverage at all nominal levels in the pooled test (Table IV). The bands still under-cover. From calibration to test, base coverage fell from 50.1% to 37.8% for the nominal 50% band, from 78.6% to 72.1% for 80%, and from 87.5% to 84.6% for 90%. Test bands also narrowed, consistent with a shift in the relation between errors and interval widths. Adjacent forecasts share dates and overlapping multi-day targets, weakening exchangeability. Monthly rolling calibration improved on base coverage but lagged static calibration at each level. The supplement gives coverage and width by lead. We assess width and misses together with the interval score [26]. For nominal miscoverage α, it adds $2/\alpha$ times the distance outside the band to its width. Static calibration had the lowest pooled score at all levels (Table IV). Rolling improved on the base score at each level.

C. RQ2: Policy choice and the cargo objective

Table III compares decisions against the robust incumbent. R1 reduced mean synthetic cargo cost regret from USD 9,603 to USD 482, a reduction of USD 9,121. Mean time regret increased by 70.75 h. The identical Wilcoxon p-values (time $p = 1.7 \times 10^{-10}$, cost $p = 1.7 \times 10^{-10}$) reflect the same 41 changed choices: 41 increases and 0 decreases in time, and 41 decreases and 0 increases in cost. It captured 41 of 42 beneficial cargo interventions, and 0 of its 41 assessable interventions were unnecessary (0.0%). R2 chose an option in 111 cases and had zero cost regret on that subset. With either fallback, R2 matched R1 and R5 on all 182/182 cargo choices. We report a single cost-objective result for those rules. Time-only and the incumbent chose identically on all 182 cargo cases, so the cargo-blind comparison is equivalent to R1 versus the incumbent. The time-only rule cost more than R1 in 22.5% of pairs. In those cases the time-only choice cost USD 40,487 more on average (95% CI 33,841–47,057). The supplement plots the mean time–cost frontier. Vessel R5 captured 0 of 10 beneficial test interventions, below the gate's 50% capture requirement. It failed the validation capture check as well.

D. RQ2: Robustness to hidden outcomes

With decisions held fixed, R1 retained 79.4–104.1% of its nominal mean cargo gain under the closure, dependence and fuel perturbations detailed in the supplement.

E. RQ2: Tail behaviour of the risk-averse rules

Predictive CVaR, the R3 objective, averages the upper tail of simulated total costs for one option in one decision. Realised CVaR in Table V averages the upper tail of the chosen options' realised total costs across cases, with fractional weight at the cutoff. We report maximum regret as a point value with its case identifier [27]. The supplement gives intervals for the other tail metrics.

Against the incumbent, R3 and R4 were worse on both observed means. Mean time regret rose by 82.88 and 87.55 h, and mean cost regret rose by USD 9.18k and USD 17.05k. The 95% cost interval includes zero for R3 [-4.00, 24.78] and excludes zero for R4 [1.16, 35.70], in USD thousands. Their benefit was fewer regrets above USD 10,000 (4.4% and 5.5% versus 20.9%), with the largest single losses.

One synthetic case, cargo-5136, gave R3 and R4 the same USD 759,546 maximum regret. Both diverted via the Cape

TABLE III
SYNTHETIC CARGO ABLATION. MEAN REGRET DIFFERENCES VERSUS THE ROBUST INCUMBENT ON CASES WITH OUTCOMES FOR BOTH THE RULE AND THE INCUMBENT. BRACKETS SHOW PAIRED 95% INTERVALS. LOWER IS BETTER.

Rule	Pairs	Δ time regret (h)	Δ cost regret (USD thousands)
Time only	182	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
R1 (= R2 with fallback = R5; identical choices 182/182)	182	70.75 [51.07, 91.07]	-9.12 [-12.05, -6.37]
R2, conditional	111	116.00 [86.27, 146.69]	-14.95 [-19.45, -10.73]
R3	182	82.88 [62.17, 104.63]	9.18 [-4.00, 24.78]
R4	182	87.55 [66.29, 109.52]	17.05 [1.16, 35.70]

Time-only and the incumbent chose identically in all 182 cargo cases. The cargo-blind comparison equals R1 versus the incumbent.

TABLE IV
POOLED FORECAST INTERVAL PERFORMANCE ON 4,260 FORECASTS. WIDTH AND SCORE ARE IN INDEX POINTS.

Nominal	Method	Coverage (%)	Width	Score
50%	Base	37.8	7.58	21.74
	Static	38.9	7.96	21.64
	Rolling	38.6	7.84	21.67
80%	Base	72.1	18.03	31.53
	Static	75.1	19.32	31.05
	Rolling	73.8	18.76	31.27
90%	Base	84.6	23.88	38.12
	Static	89.4	26.82	37.23
	Rolling	86.9	25.18	37.92

and incurred large late penalties. The model assigned 29.9% probability to at least that much regret for the diversion against the cheapest action in each simulated future. The model anticipated an outcome of this size, and the realised chosen cost stayed below both predicted VaRs. The case contributed 17.3% and 15.9% of R3's and R4's realised CVaR, respectively. Both still have higher realised CVaR than R1 when that case is excluded. Both also raised the frequency of regret above the threshold in the vessel corpus. Mean vessel time regret was 1.05 h for the incumbent and R1, 30.28 h for R3 and 22.17 h for R4.

Extending the fallback tails changed 1 R3 choice and left R1 and R4 unchanged. Extending the forecast pressure tails left choices unchanged for all three rules in both forecast arms.

VaR exceedance means realised cost above the chosen option's predicted quantile. Expected counts $N(1-\alpha)$ at 90% and 95% are about 18/9 for 182 cases. Observed counts were 15/9 (R1 group), 16/9 (R3), 15/8 (R4), and 17/9 (time only). Conditional R2 had 7/4 against about 11/6 expected in 111 cases. The Kupiec tests [28] have low power at these counts, so non-rejection is weak evidence of calibration. Both backtests and exceedance sizes are in the supplement.

F. RQ3: Do forecast changes reach decisions?

R1 switched 0/182 cargo choices and 0/4 fixture choices. Mean cost regret stayed the same because all pairs retained their choices and shared the same scoring outcomes. The other tested rules also kept their choices, including R2 with its keep fallback.

We isolate each option's expected wait cost by removing berth exposure and repricing the same futures. We divide the largest absolute change in this cost across options by the original R1 gap to the runner-up. Across the 181 cases with positive margins, the median margin was USD 21,371. The ratio's median, p90 and maximum were 0.04%, 0.65%, and 23.3%. Median original wait and cargo shares were 0.3% and 67.8% across options. The corpus centres on chokepoint closures that can delay voyages by days, while the illustrative berth-wait mapping changes arrivals by hours.

Symmetric widening about a fixed median preserves the unclipped linear wait mean. Asymmetry, pressure bounds and the zero floor can shift it. At triple width, raw mean wait changed by -0.30 h and the floor added 1.99 h before exposure. Removing exposure on matched draws showed median absolute changes across 612 options of USD 167 in expected wait cost, USD 729 in its upper-tail mean and USD 270 in wait's contribution to total-cost CVaR. The supplement gives all cells.

We multiply the calibrated half-widths by k around the fixed median and the wait slope by m, bound pressure to its scale and retain the same uniforms and outcomes. R1 stayed fixed throughout the grid. By increasing km and breaking ties by smaller m, R3 first reached 1% switches at $k = 3$, $m = 1$ (2/182). Neither rule reached 5%. The supplement gives the full grid and paired mean cost-regret intervals.

G. Engine precision and case studies

For the Suez reefer fixture, the first stable tier required 4,096 futures per option with independent pseudorandom draws and 512 with shared pseudorandom or shared scrambled Sobol draws. Stability required two consecutive same-sign 95% intervals excluding zero for a pilot-selected option pair. At 65,536 draws, the independent SE was 11.7 times the shared Sobol SE. The supplement plots convergence.

The supplement shows how the keep–Cape cost gap varies with closure duration and remaining shelf life. The time-critical Suez fixture flips from keep to Cape in the 229.5–243.0 h closure bracket. The S10 passage fixture favours slower steaming at the same conditional approach time. Both synthetic Suez mixes and the illustrative Ever Given fixture favour keep in expected cost. The latter uses a recorded reopening only in conditional scoring [29].

VIII. DISCUSSION

The cargo cost gain came from pricing the cargo objective. R1 and R5 chose identical options, and the hybrid failed the

TABLE V
SYNTHETIC COST TAILS IN USD. CVAR AND VAR USE 95%. R1, R2 WITH EITHER FALLBACK AND R5 SHARE THE R1 GROUP ON CARGO. DASHES IN THE VAR COLUMNS INDICATE NO BACKTEST.

Rule	Regret above 10,000 (%)	p90 regret	Worst regret (case)	Realised CVaR	VaR exceed. (%)	Kupiec p
Cargo						
Incumbent	20.9	44,395	89,251 (cargo-5139)	346,464	–	–
Time only	20.9	44,395	89,251 (cargo-5139)	346,464	4.9	0.973
R1 group	0.5	0	87,741 (cargo-5198)	346,464	4.9	0.973
R2 conditional[a]	0.0	0	0 (—)	128,526	3.6	0.478
R3	4.4	0	759,546 (cargo-5136)	572,489	4.9	0.973
R4	5.5	0	759,546 (cargo-5136)	620,786	4.4	0.703
Vessel						
Incumbent	1.6	0	16,850 (vessel-5015)	72,409	–	–
R1	1.6	0	16,850 (vessel-5015)	72,409	–	–
R3	13.2	39,512	104,250 (vessel-5046)	92,547	–	–
R4	10.5	19,292	89,175 (vessel-5076)	92,447	–	–

[a] Conditional R2 had zero regret on its native subset of cases where it selected an option. On cargo, time-only and the incumbent chose identically in all 182 cases, so the cargo-blind comparison equals R1 versus the incumbent. R1 and the incumbent chose keep on all 10 cases contributing to the same realised CVaR, including the fractionally weighted cutoff case.

vessel promotion test. The lower cargo regret came with more time regret. A planner can use the cost ranking to identify which delivery or insurance terms justify extra sailing time and review that trade-off for each option.

R3 and R4 were worse on both observed means than the incumbent (+82.88 and +87.55 h, and +USD 9.18k and +USD 17.05k). The cost interval spans zero for R3 and excludes it for R4. Both rules reduced regrets above USD 10,000 (4.4% and 5.5% versus 20.9%) and shared the largest loss, which the model had assigned 29.9% probability of reaching or exceeding.

In this closure-centred corpus, delays can span days, while the wait map moves arrivals by hours. Median wait and cargo shares were 0.3% and 67.8%. Recalibration changed wait costs by small fractions of the original choice margins. Widening the pressure band around a fixed median had more effect on upper-tail wait costs than on mean wait costs. Tail-sensitive rules can respond when those tails move, as R3 did in the stress grid. Steeper wait slopes can amplify the cost effect, and a biased median could move the expected-cost ranking.

Interactive sampling can resolve large gaps quickly. For the synthetic Suez stress case, reaching the USD 25 paired standard-error target took 524,288–1,048,576 futures and 4.4–7.9 s. A close ranking or a tail-sensitive choice can warrant a longer offline run.

IX. THREATS TO VALIDITY

The reporting period reuses previously examined dates. PortWatch reports arrive with a lag and can be revised [2]. Target dates approximate availability because publication vintages are incomplete.

The overlay pairs forecast rows with unrelated synthetic voyage ports and dates. Days of closure delay and hours of mapped berth wait make wait a small cost term. The zero-switch finding applies to this corpus and mapping. The vessel corpus uses an illustrative USD 250 per delay-hour. Historical fixtures use illustrative hull and cargo inputs.

The hidden outcomes share assumptions with the simulator. Stress tests vary selected parameters, dependence and one alternative to clamping, which puts mass at the endpoints and can affect predicted tails. Extreme outcomes have limited empirical support. The grid holds the median fixed. A biased median could change expected wait and R1 rankings.

X. CONCLUSION AND FUTURE WORK

Static conformal recalibration improved empirical coverage, though bands still under-covered. Causal regime features did not improve the two prespecified forecast metrics jointly. Ordinary recalibration left the synthetic choices unchanged. Expected-cost selection lowered synthetic cost regret and increased time regret.

Future work includes regime-driven Monte Carlo, a sequential voyage reinforcement-learning environment, scene-based SAR and optical ship detection, and a held-out test of source specialists on hand-labelled inputs. Operational validation also needs timestamped forecasts and verified arrivals, berth waits and cargo outcomes.

DATA AND CODE AVAILABILITY

The repository contains the frozen panel, code, manifest, and numeric ledgers. The companion supplement is `docs/paper/supplement.pdf`.
https://github.com/RaghhavMalani/logistics-route-optimization

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